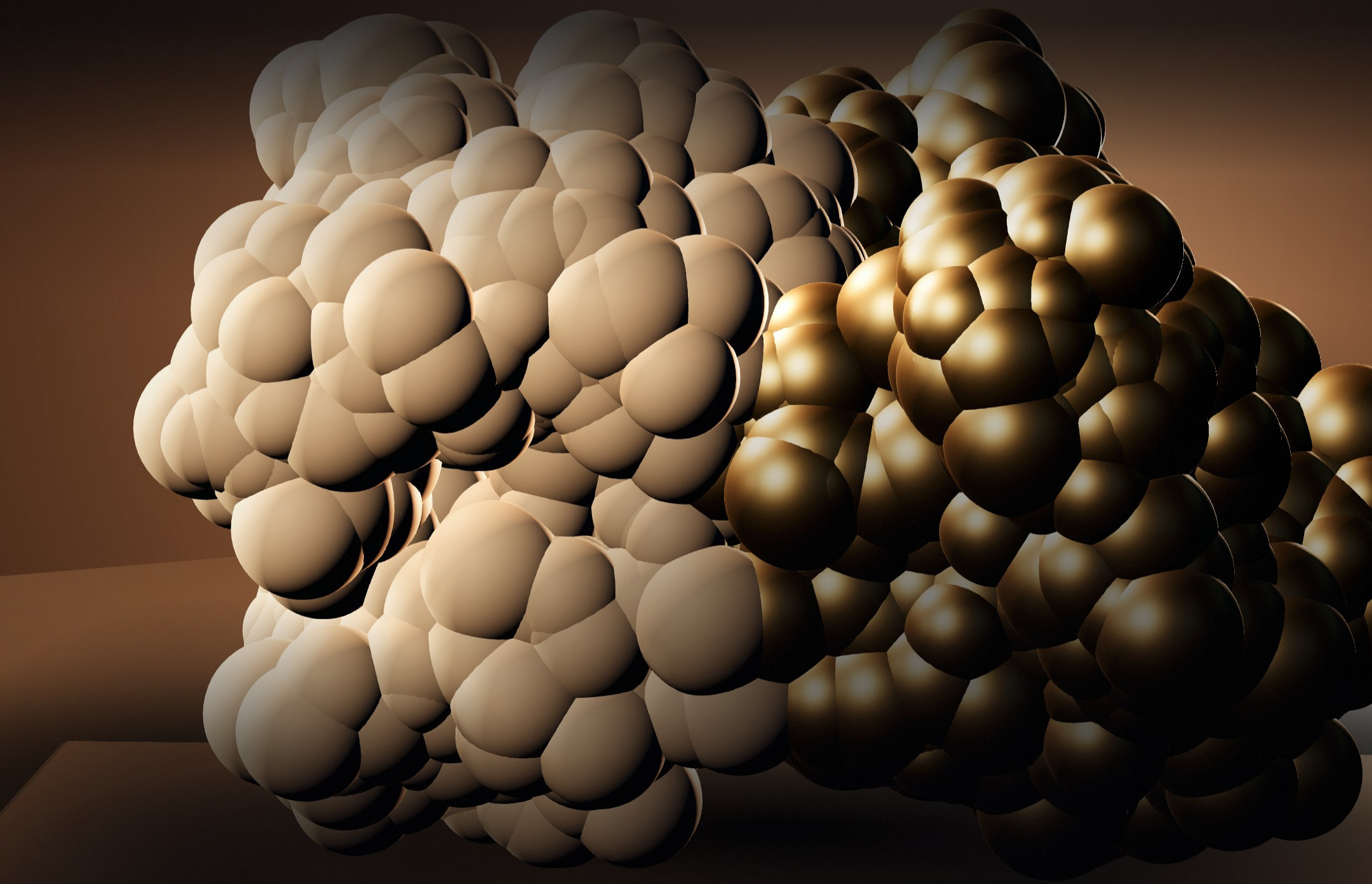
An AI was asked which proteins pair up

Texas A&M scientists in College Station recast AlphaFold-Multimer as a yes or no test for which mitochondrial proteins interact. The model is drawn to illustrate, not a structure from the paper.



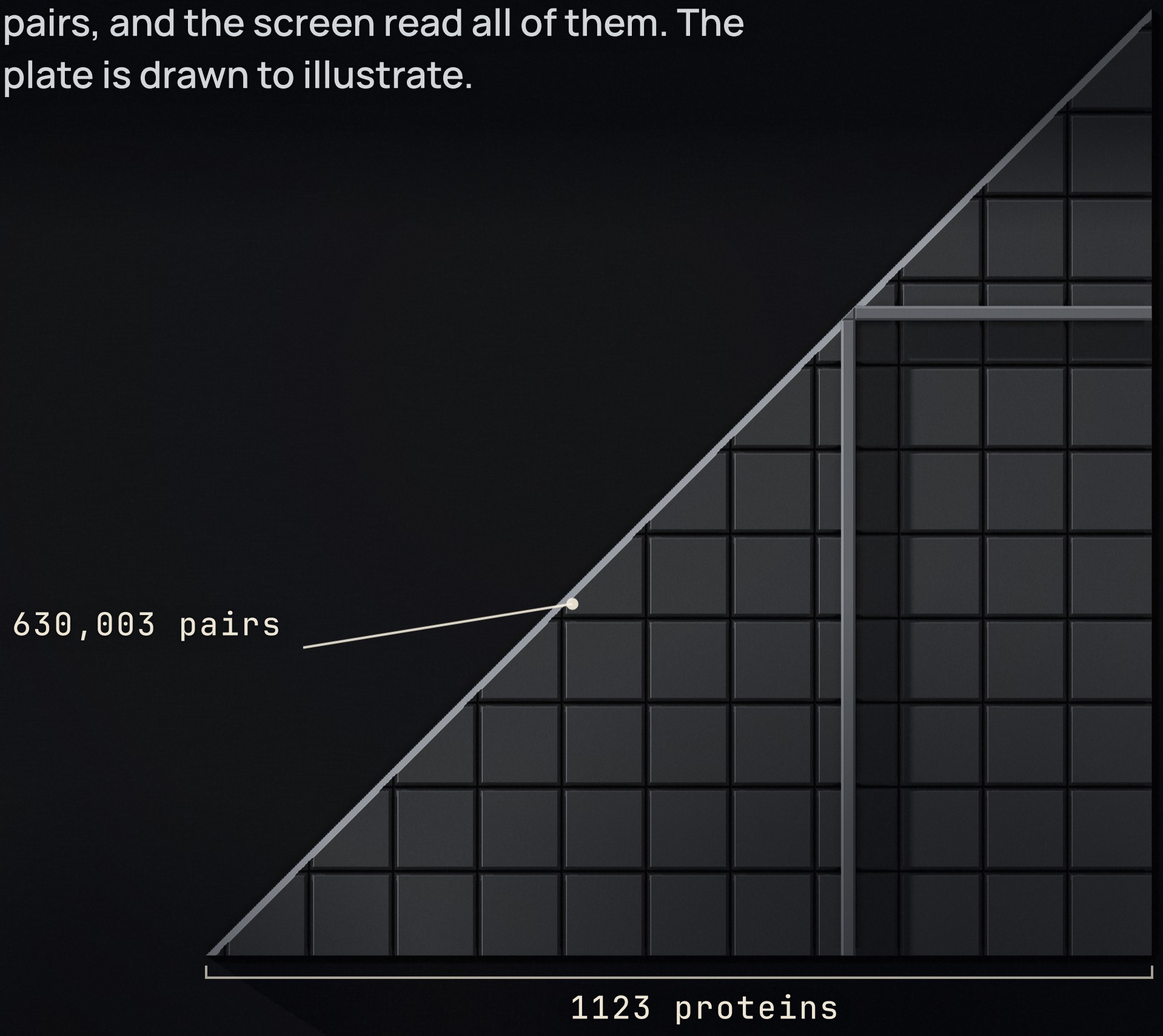
Do these two meet?

Google DeepMind built AlphaFold-Multimer to predict how protein chains fit together. The team used it as a yes or no test on each pair. Mutations in nearly 30% of mitochondrial proteins are tied to disease. The models are drawn to illustrate.



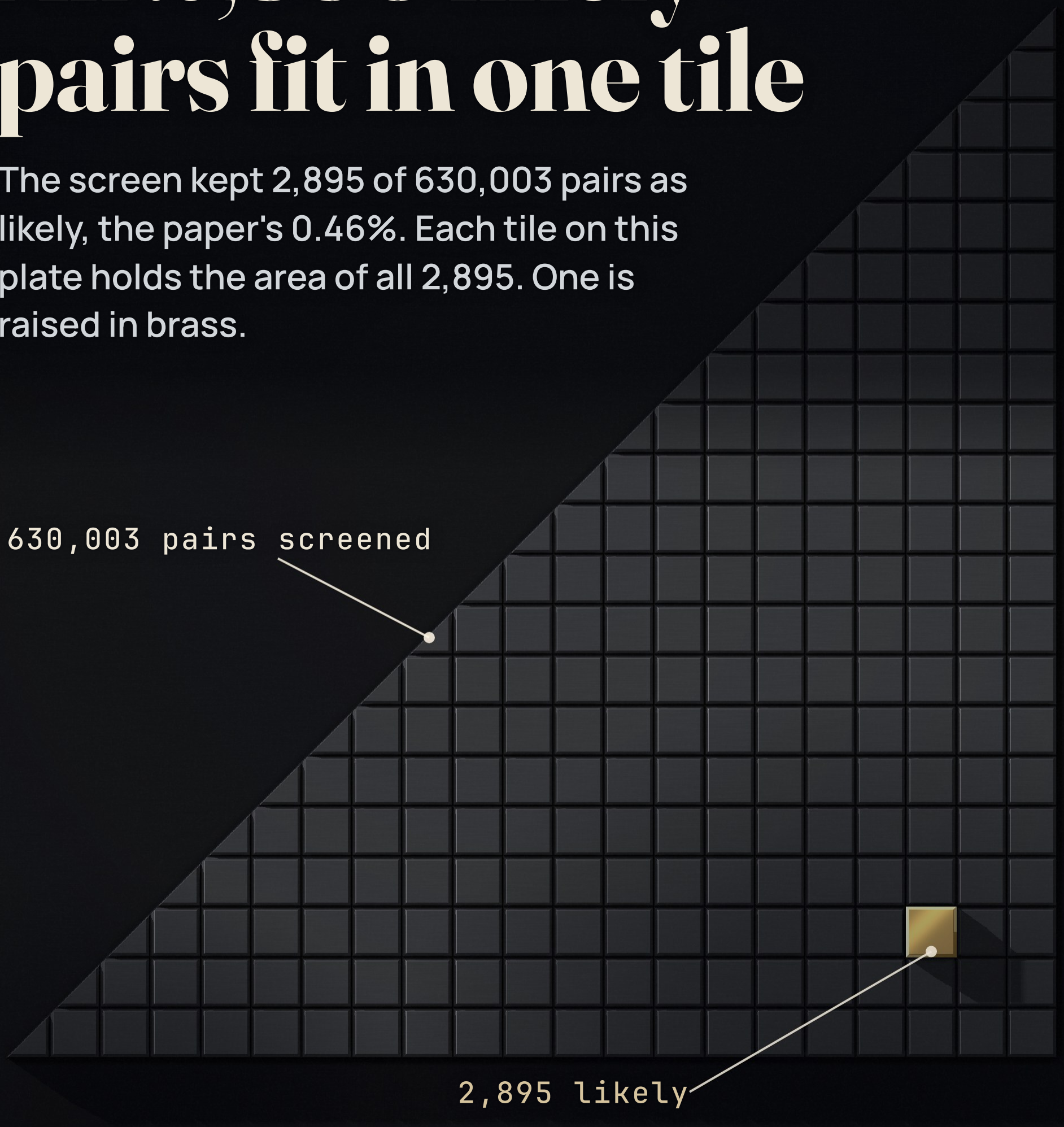
Every pair, once

The team built a curated list of 1123 proteins. Every possible pairing comes to 630,003 pairs, and the screen read all of them. The plate is drawn to illustrate.



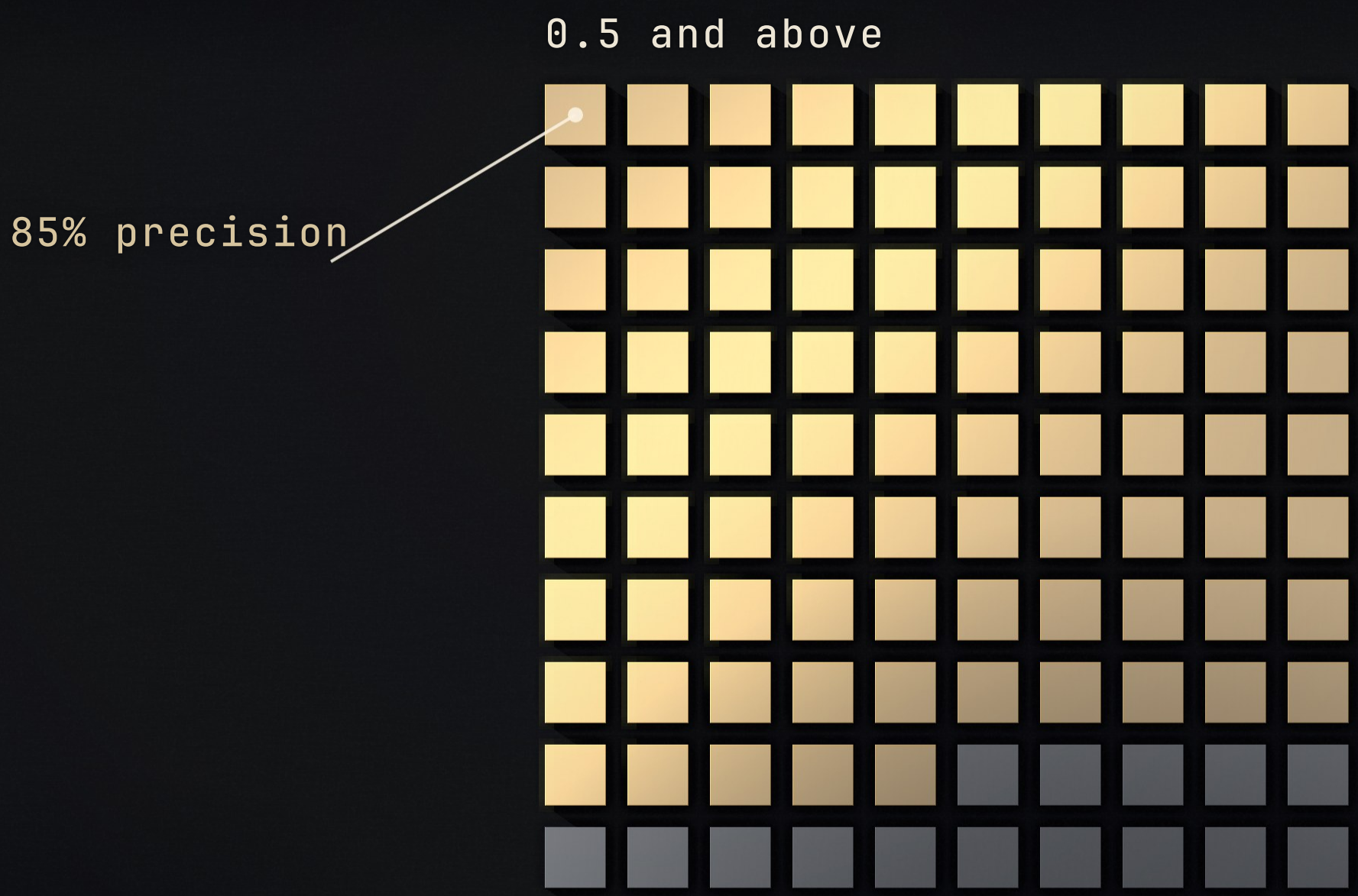
All 2,895 likely pairs fit in one tile

The screen kept 2,895 of 630,003 pairs as likely, the paper's 0.46%. Each tile on this plate holds the area of all 2,895. One is raised in brass.



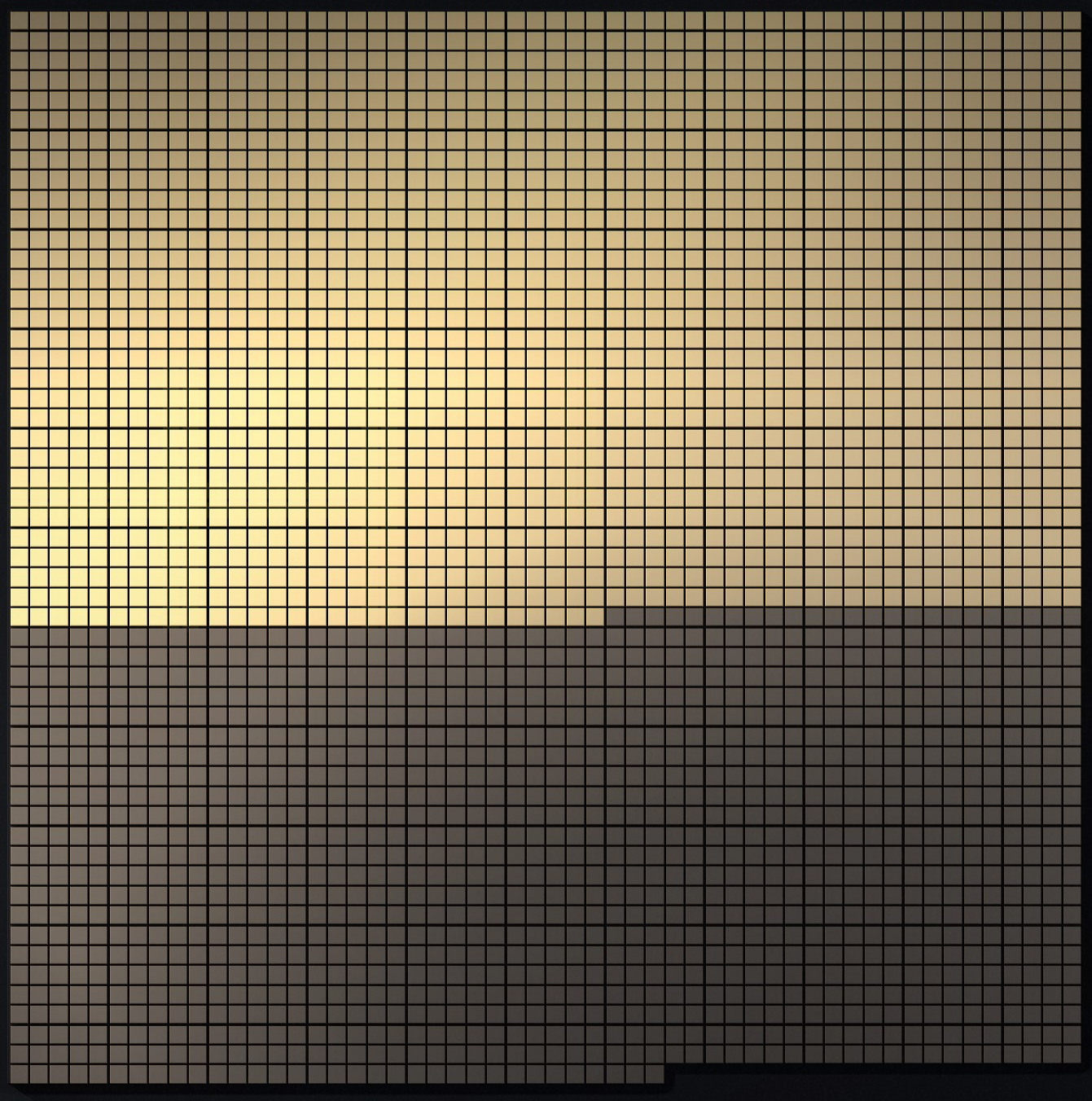
The line sits at 0.5

A pair counts when its mean ipTM score reaches 0.5. On the authors' test set that gives 67% recall at a false discovery rate of about 15%. Of a hundred calls drawn above the line, the 15 in steel are the false share.



Most of them had never been reported

57% of the predicted interactions were in none of the databases checked. Every cell stands at one height, because the method can't say how tightly a pair binds.

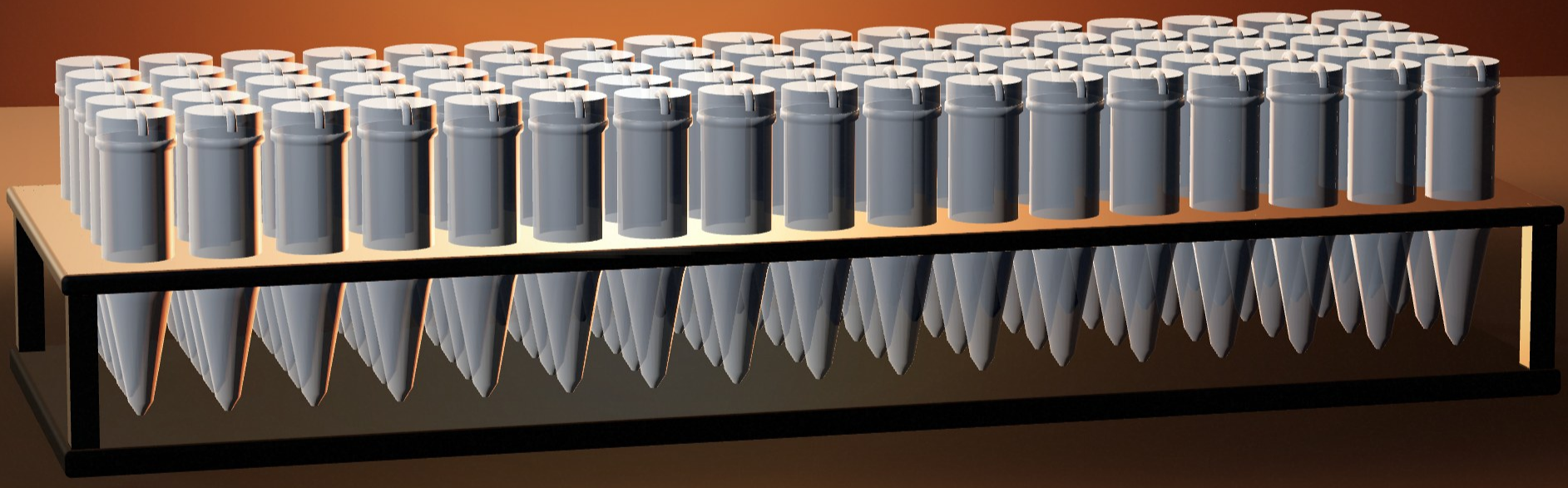


c13 c14 c24

texasaidocket.com

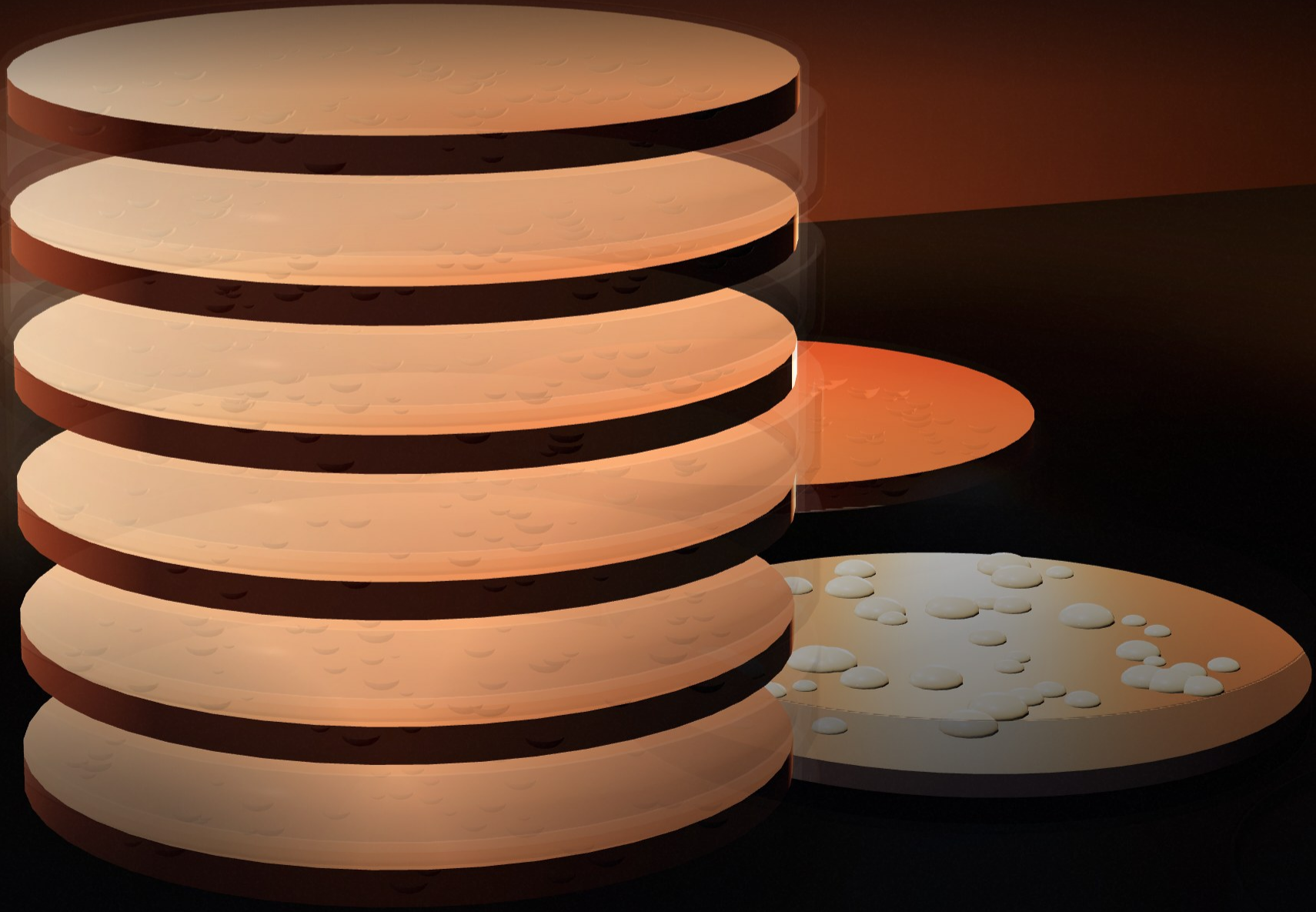
85 little-known proteins got a predicted partner

The screen found at least one predicted partner for each of 85 uncharacterized proteins. One tube stands for each, drawn to illustrate.



Several pairs held up at the bench

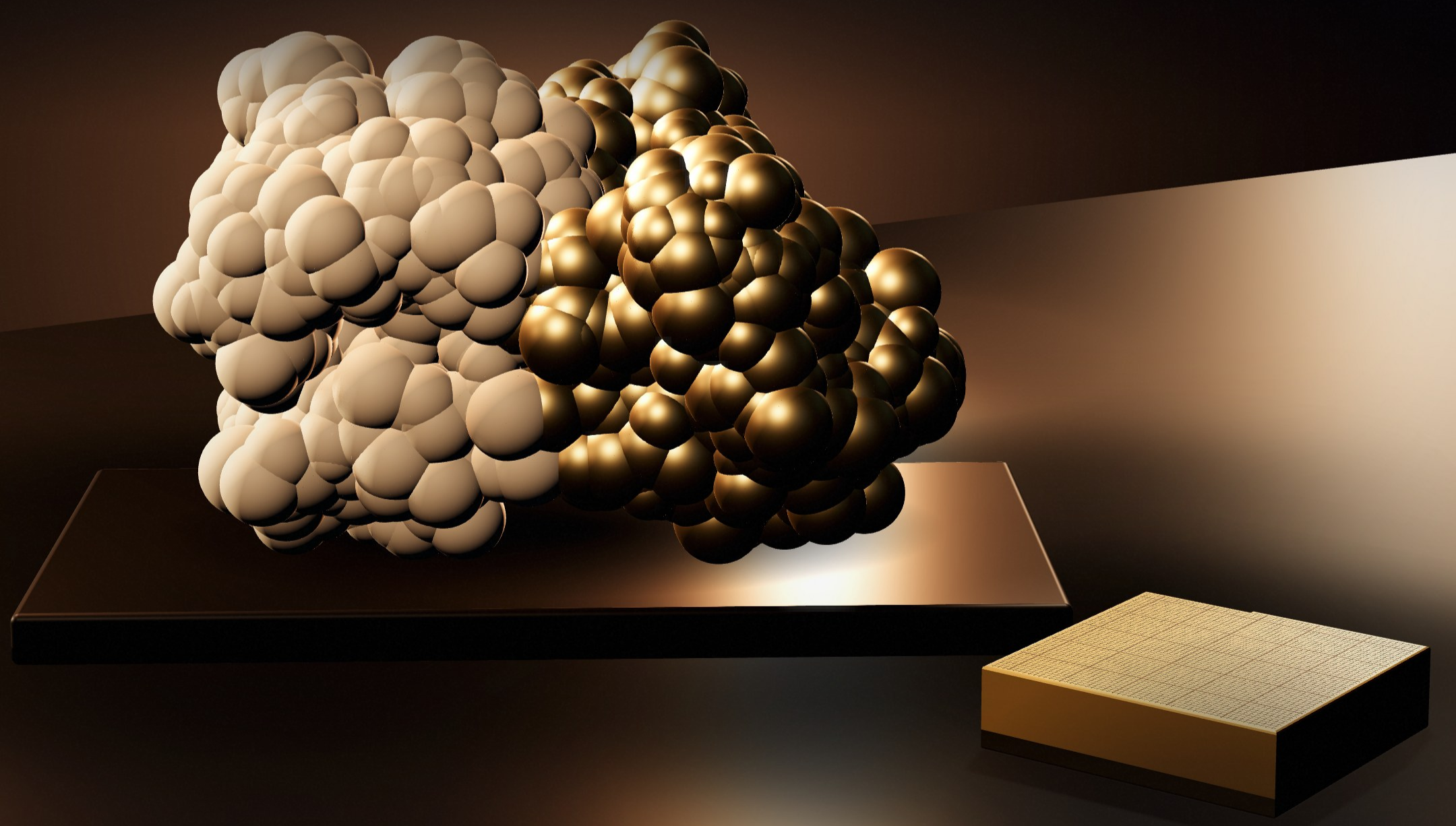
About two years of lab work confirmed several predicted partnerships in yeast and human cells. The work also mapped the route copper takes inside mitochondria to cytochrome c oxidase. The plates are drawn to illustrate.



c19 c18

The list is free to use

The team posted all 2,895 predictions as MitoMatch, free at mitomatch.web.app. The release says cures for diseases linked to mitochondria may still be far off. The block and model are drawn to illustrate.



c4 c20 c21 c23 c29